On the Performance of Single Side-Band OFDM for Band-Limited Visible Light Communication

Hossein Kazemi and Harald Haas
LiFi Research and Development Center, Institute for Digital Communications, School of Engineering,
University of Edinburgh, King's Buildings, Edinburgh, EH9 3FD, UK.
Email: {h.kazemi, h.haas}@ed.ac.uk

***Abstract*—The use of a multicarrier modulation scheme based on orthogonal frequency division multiplexing (OFDM) is a very promising approach to enable high-speed transmissions for band-limited visible light communication (VLC) systems. The existing variants of optical OFDM have been designed mostly to establish a compromise between spectral and power efficiencies, with the premise that the signal is purely transmitted in the baseband, thereby it needs to be real-valued. A common way to synthesize a real waveform with the digital implementation of OFDM is to use Hermitian symmetry. An alternative, but not fully explored, solution without imposing Hermitian symmetry is to upconvert the complex baseband OFDM signal within the limited bandwidth of VLC systems using an appropriate carrier frequency. In this paper, the bandpass transmission of OFDM signals based on optical single side-band OFDM (SSB-OFDM) is considered. The detailed mechanism for the generation and detection of an SSB-OFDM signal with discrete-time processing is presented. To address the bandwidth constraint, spectrum shaping of the SSB-OFDM signal is discussed. Furthermore, performance analyses of spectral efficiency (SE) and bit error ratio (BER) are carried out and the results are verified using Monte-Carlo simulations. Novel insights are provided, highlighting the signal-to-noise ratio (SNR) gain of SSB-OFDM over a baseline direct current-biased optical OFDM (DCO-OFDM) system.**

***Index Terms*—Single side-band orthogonal frequency division multiplexing (SSB-OFDM), visible light communication (VLC), bandwidth limitation, DC-biased optical OFDM (DCO-OFDM).**

I. Introduction

Most visible light communication (VLC) systems operate based on intensity modulation and direct detection (IM-DD), in that the information-bearing signal modulates the intensity of a light emitting diode (LED), and the variations are detected by a photodiode (PD) at the receiver. The frequency response of VLC systems is largely influenced by front-end device filtering, originated from the finite modulation bandwidth of LEDs and PDs [1]. In practice, the bandwidth limitation is primarily imposed by the cut-off frequency of LEDs. For low-cost white LEDs, the modulation bandwidth is in the order of tens of MHz when blue filtering is applied [2].

An effective approach to unlock high data rates of Gbits/s in band-limited VLC systems relies on rate-adaptive multicarrier modulation schemes, in particular orthogonal frequency division multiplexing (OFDM) [3]. The application of OFDM for indoor VLC systems offers many advantages including: 1) The power spectrum of an OFDM signal entails sharp transition bands for a sufficiently large number of subcarriers, which means that the transmitter can be matched to the low-pass characteristics of LEDs; 2) Considering a typically small root mean square delay spread of time dispersive indoor VLC channels, their frequency selectivity effectively turns flat over every subcarrier bandwidth by appending a very small cyclic prefix (CP). Thus, a single-tap equalizer suffices to create an inherent robustness for the signal against multipath-induced inter-symbol interference (ISI) [2]; 3) Baseband modulation and demodulation can be efficiently implemented by using digital signal processing (DSP) platforms providing inverse fast Fourier transform (IFFT) and fast Fourier transform (FFT).

An OFDM signal is realized by applying IFFT to a block of symbols drawn from a digital constellation diagram such as M-ary quadrature amplitude modulation (QAM). The IFFT operation maps M-QAM symbols to different subcarriers in the frequency domain resulting in complex-valued samples in the time domain. However, optical IM-DD systems constrain the baseband waveform to be solely real-valued. By enforcing *Hermitian symmetry* on the information block, the IFFT operation generates real samples. The output signal is still bipolar, while the optical intensity is a non-negative quantity in nature. A commonly used technique to produce a unipolar OFDM signal is direct current-biased optical OFDM (DCO-OFDM), in which the varying component of the signal is shifted to a positive level by choosing a proper value for DC bias [4]. Due to the high peak-to-average power of OFDM, a large DC bias is usually required, which makes DCO-OFDM inefficient in terms of the electrical power dissipation. An alternative method is to hard-clip the entire negative excursion of the signal so that it can be sent through IM-DD channels without the need for DC bias. In order to avoid clipping noise, methods such as asymmetrically clipped optical OFDM (ACO-OFDM) are used. In ACO-OFDM, only odd subcarriers are modulated, whereas the clipping noise frequency components fall on even subcarriers which are not used for data transmission [5]. These techniques are mainly intended for *baseband* communications in optical IM-DD systems.

In radio frequency (RF) wireless communication systems, the baseband OFDM signal is complex-valued. The real and imaginary parts of this complex signal are upconverted with an RF carrier using a quadrature carrier modulator, resulting in a real OFDM signal operating in the *passband* around the carrier frequency. The RF upconversion shifts the signal spectrum to frequencies much higher than the signal bandwidth. When it comes to VLC systems with off-the-shelf LEDs, the bandwidth limitation makes it challenging to upconvert OFDM signals from the baseband to passband. A spectrum efficient approach

978-1-7281-7440-2/20/$31.00 ©2020 IEEE
Authorized licensed use limited to: University of Wollongong. Downloaded on August 14,2020 at 14:36:50 UTC from IEEE Xplore. Restrictions apply.

to cope with this problem is based on single-side band OFDM (SSB-OFDM), whereby only one of the lower or upper side-bands of a baseband OFDM signal, i.e. either LSSB-OFDM or USSB-OFDM, is transmitted. Although SSB-OFDM has been widely adopted in optical fiber communications for compensating the chromatic dispersion in long-haul fiber links, e.g. [6], [7], there are a few studies for VLC applications, e.g. [8]–[10]. In [8], Kim *et al.* have proposed a carrier allocation technique based on SSB-OFDM to mitigate the inter-cell interference in broadband VLC networks. The authors have also investigated their proposed technique for indoor visible light positioning by employing a separate designated frequency for each LED lamp, in [9]. Moreover, the work in [10] presented an experimental verification of SSB-OFDM with bit and power loading for carrier allocated VLC systems. The main incentive for carrier allocation, however, is to modulate multiple signals within the system bandwidth. Still, more fundamental questions of how SSB-OFDM operates for a standalone VLC system and what potential benefits it offers over the existing OFDM-based solutions remain unanswered.

This paper attempts to address the aforementioned questions and shed light on such aspects using a theoretical approach. More specifically, the transceiver architecture of SSB-OFDM with DC bias is presented and the associated signal processing steps are elucidated. The spectral efficiency (SE), received electrical signal-to-noise ratio (SNR) and bit error ratio (BER) performances of SSB-OFDM for a band-limited VLC system are derived and compared with DCO-OFDM. The paper begins with an overview of the baseline DCO-OFDM system.

II. Baseband Optical OFDM System

Most of the commercially available LEDs are shown to have a flat frequency response for modulation bandwidths of about 2–20 MHz [11]. This allows to safely assume a flat channel for the entire OFDM frame over the limited bandwidth of VLC systems. In addition, for indoor VLC channels deployed in typical office environments where maximum delay spreads are up to 100 ns, the ISI caused can be fully compensated by using a CP composed of only two samples [2]. In this paper, the overhead associated with CP is not considered to simplify the system-level study, and DCO-OFDM is used as a benchmark for comparison. Fig. 1 depicts a conventional DCO-OFDM transceiver based on an N-point IFFT/FFT for optical IM-DD systems. As the signal progresses through different stages, the function of each stage is described concisely in the following.

From Fig. 1, observe that the transformed data symbols are directly sent over the channel without any further modulation or upconversion. This means it is only possible to transmit a real signal in the time domain, so the IFFT output is restricted to be real-valued. A common way to ensure such a real output is by applying Hermitian symmetry in the frequency domain. To this end, the double-sided baseband frequency span of the system from $-B$ to $+B$ is divided into N subchannels, with each one acquiring an equal bandwidth of $\frac{2B}{N}$. Based on an M-QAM modulation, with a constellation of size of $M = 2^b$, the input binary information sequence is arranged

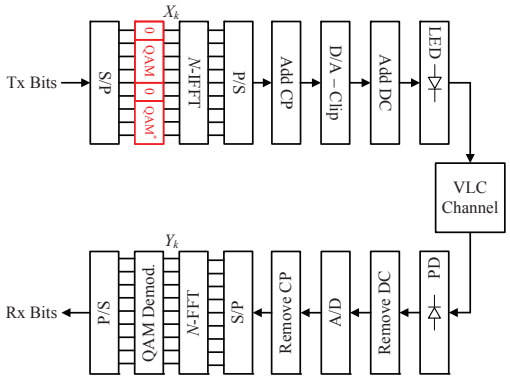


Fig. 1: Block diagram of a DCO-OFDM transceiver for optical IM/DD systems.

in a parallel format and buffered into blocks of b bits to be loaded onto $N_{\mathrm{QAM}} = \frac{N}{2} - 1$ data-carrying subcarriers. The bits assigned to the kth subcarrier are mapped to the kth QAM symbol denoted by X_k. The average power of QAM symbols are normalized to unity such that $\mathbb{E}\left[|X_k|^2\right] = 1$, where $\mathbb{E}$ denotes the expectation operator. The total number of symbols is extended to N by imposing a Hermitian symmetry so that $X_{N-k} = X_k^*$ for $k = 1, 2, \ldots, N_{\mathrm{QAM}}$, where * denotes complex conjugate. To maintain the conjugate symmetry, it is essential that X_0 and $X_{\frac{N}{2}}$ be real. Since X_0 is reserved for DC bias, the 0th and the π-shifted subcarriers are left unmodulated by setting $X_0 = X_{\frac{N}{2}} = 0$ [2], as shown in Fig. 1. This leads to a subcarrier utilization ratio of $\xi_{\mathrm{DCO}} = \frac{N-2}{N}$. The resulting DCO-OFDM frame has the following structure:

$$\mathbf{X} = \left[0, X_1, X_2, \ldots, X_{N_{\mathrm{QAM}}}, 0, X^*_{N_{\mathrm{QAM}}}, \ldots, X^*_2, X^*_1\right]. \quad (1)$$

Applying an N-point IFFT transforms the elements of $\mathbf{X}$ into purely real samples in the time domain. The IFFT output at time sample n, for $n = 0, 1, \cdots, N-1$, is given by:

$$x(n) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_k e^{j\frac{2\pi kn}{N}}, \quad (2)$$

where $j = \sqrt{-1}$. Subsequently, the N samples are converted from the parallel to serial (P/S) format and a CP is appended to the beginning of the OFDM symbol. For $N \geq 64$, the IFFT output signal in (2) follows a zero mean and real Gaussian distribution based on the central limit theorem [12]. In order to establish an average electrical power of P_{elec} for OFDM symbol, the signal variance is first normalized to unity and then scaled by a factor equal to $\sqrt{P_{\mathrm{elec}}}$. Finally, after digital-to-analog (D/A) conversion, the desired DC bias is added and the remaining negative samples are clipped at zero. The resulting unipolar waveform is used to drive the forward current of LED. Assuming a lossless electrical-to-optical conversion over the operating linear dynamic range of LED, the continuous-time output optical power at time t can be written as follows:

$$x_{\mathrm{DCO}}(t) = \sqrt{\frac{P_{\mathrm{elec}}}{\xi_{\mathrm{DCO}}}} x(t) + x_{\mathrm{DC}}, \quad (3)$$

where x_{DC} is the DC bias. This signal represents instantaneous variations of the light intensity subject to the nonlinear current-

Authorized licensed use limited to: University of Wollongong. Downloaded on August 14,2020 at 14:36:50 UTC from IEEE Xplore. Restrictions apply.

to-power transfer characteristic of LED. In practice, prior to DC addition, the modulating signal is symmetrically clipped from top and bottom in consideration of the limited dynamic range of LED. A reasonable choice for DC bias is one that sets the average optical power, $P_{\text{opt}} = x_{\text{DC}}$, at the midpoint of the LED dynamic range, and that satisfies $x_{\text{DC}} = 3\sqrt{P_{\text{elec}}}$. The latter condition guarantees that 99.7% of the signal envelope remains undistorted, as 99.7% of the optical power is confined within the range $(-3\sqrt{P_{\text{elec}}}, +3\sqrt{P_{\text{elec}}})$, and as a result, the clipping noise can be effectively discarded [13].

At the receiver, a PD transduces the instantaneous optical power into an electrical photocurrent. The DC component is then subtracted from the detected photocurrent. Followed by analog-to-digital (A/D) conversion, and the removal of CP, the resulting N output samples are arranged in a parallel format. By performing an N-point FFT, the N_{QAM} complex M-QAM symbols are extracted from their respective subcarriers. The symbols transmitted on different subcarriers are individually demodulated and the decoded data bits are rearranged from the serial to parallel (S/P) format.

III. Single-Side Band OFDM for Band-Limited VLC

The key idea is to transmit an LSSB-OFDM signal whose main side-band in the frequency domain effectively lies within the single-sided bandwidth of a VLC system. This way, the low pass filter (LPF) effect only influences the upper side-band which carries no information anyway, and therefore the signal spectrum is not distorted. A bandpass SSB-OFDM signal can be generated by means of a quadrature carrier modulator when the real and imaginary components of the baseband OFDM signal are related by a *Hilbert transform*. They are used as the drive signals for the in-phase and quadrature mixers, respectively [14]. This principal is combined with the proposed idea to devise an SSB-OFDM transceiver architecture suitable for band-limited optical IM-DD channels. The detailed block diagram is illustrated in Fig. 2. The design follows a digital implementation using N-point IFFT/FFT operations similar to the one in baseband optical OFDM systems, as shown in Fig. 1. There are additional blocks in Fig. 2 as highlighted with a red color for comparison with Fig. 1. Note that SSB-OFDM has a bipolar waveform and its conversion to a unipolar signal can be achieved using DC biasing, as shown in Fig. 2.

A Hilbert transform can be applied at the OFDM transmitter by setting the negative subcarriers to zero [15]. Hence, unlike the case in the DCO-OFDM system, no Hermitian symmetry constraint is imposed on an SSB-OFDM frame, as shown in Fig. 2. In SSB-OFDM with DC bias, $N_{\text{QAM}} = \frac{N}{2} - 1$ symbols are arranged on the first half of the frame as follows:

$$\mathbf{X} = \left[0, X_1, X_2, \ldots, X_{N_{\text{QAM}}}, 0, 0, \ldots, 0\right], \tag{4}$$

leading to a subcarrier utilization ratio of $\xi_{\text{SSB}} = \frac{N-2}{2N}$. Note that $\xi_{\text{SSB}} = \frac{1}{2}\xi_{\text{DCO}}$, because the baseband SSB-OFDM frame takes half of the available subcarriers for data transmission. After executing an N-point IFFT as in (2) and P/S conversion, the discrete-time output signal is of the form:

$$x(n) = x_{\text{i}}(n) + jx_{\text{q}}(n), \tag{5}$$

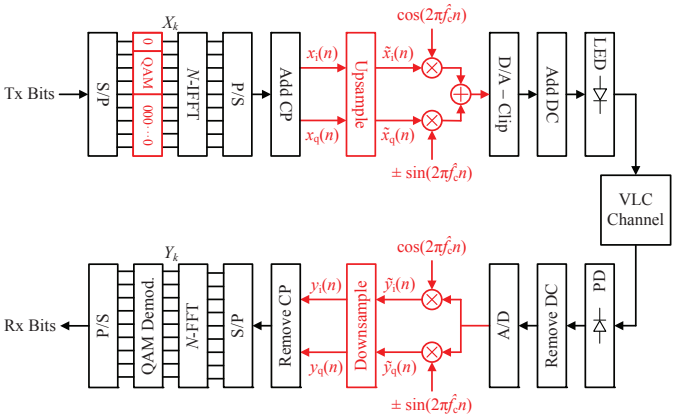


Fig. 2: SSB-OFDM transceiver architecture for band-limited optical IM/DD systems. The $\pm$ signs lead to the generation of LSSB/USSB-OFDM signals.

where $x_{\text{i}}(n) = \Re[x(n)]$ and $x_{\text{q}}(n) = \Im[x(n)]$ represent the in-phase and quadrature components, with $\Re[x(n)]$ and $\Im[x(n)]$ denoting the real and imaginary parts of $x(n)$, respectively. The signals $x_{\text{i}}(n)$ and $x_{\text{q}}(n)$ are derived as follows:

$$x_{\text{i}}(n) = \frac{1}{\sqrt{N}} \sum_{k=1}^{N_{\text{QAM}}} \Re[X_k] \cos\left(\tfrac{2\pi kn}{N}\right) - \Im[X_k] \sin\left(\tfrac{2\pi kn}{N}\right), \tag{6a}$$

$$x_{\text{q}}(n) = \frac{1}{\sqrt{N}} \sum_{k=1}^{N_{\text{QAM}}} \Re[X_k] \sin\left(\tfrac{2\pi kn}{N}\right) + \Im[X_k] \cos\left(\tfrac{2\pi kn}{N}\right). \tag{6b}$$

The Hilbert transform, denoted by $\mathcal{H}$, is equivalent to a -90° phase shift for positive frequencies and a $+90^\circ$ phase shift for negative frequencies [14]. It follows that $x_{\text{q}}(n) = \mathcal{H}[x_{\text{i}}(n)]$, since $\sin(\theta_k) = \cos(\theta_k - \frac{\pi}{2})$ and $\cos(\theta_k) = -\sin(\theta_k - \frac{\pi}{2})$, for $\theta_k = \frac{2\pi kn}{N}$.

Considering a single-sided bandwidth of B for the VLC system, the complex-valued signal in (6) has a spectrum between the frequencies $-B$ and $+B$. The Nyquist sampling frequency for this signal is given by $f_{\text{s}} = 2B$. The objective is to create an LSSB-OFDM signal that has a lower side-band of bandwidth B in the passband. For this purpose, the in-phase and quadrature signals in (6) are upsampled by a factor of two. By doing so, their frequency support is extended to the range $-2B$ to $+2B$, while keeping their *modulated* bandwidth the same as before, thereby opening up room to carry out frequency upconversion using DSP, as shown in Fig. 2. The sampling rate is increased by adding zeros between the existing samples, followed by a discrete-time LPF, which is assumed to be an ideal filter with a passband gain of 2 and an *equivalent* cut-off frequency of $\frac{\pi}{2}$ [16]. This results in an effective interpolation of the signal samples with corrected amplitudes at locations of the zeros, and the new sampling frequency becomes $2f_{\text{s}}$. The upsampled in-phase and quadrature signals are $\tilde{x}_{\text{i}}(n) = x_{\text{i}}\left(\frac{n}{2}\right)$ and $\tilde{x}_{\text{q}}(n) = x_{\text{q}}\left(\frac{n}{2}\right)$, for $n = 0, 2, 4, \ldots, 2N-2$. The amplitudes of $\tilde{x}_{\text{i}}(n)$ and $\tilde{x}_{\text{q}}(n)$ for $n = 1, 3, 5, \ldots, 2N-1$ are properly determined by the interpolation filter. The carrier frequency, denoted by f_{c}, is chosen such that the signal spectrum from $-B$ to $+B$

Authorized licensed use limited to: University of Wollongong. Downloaded on August 14,2020 at 14:36:50 UTC from IEEE Xplore. Restrictions apply.

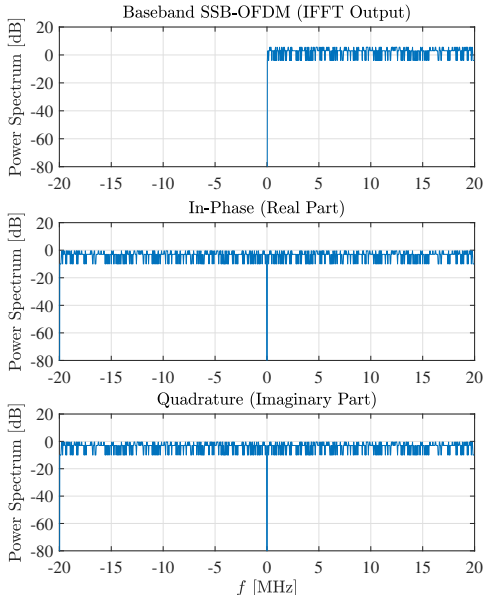


Fig. 3: Power spectra of the baseband SSB-OFDM signal and its real and imaginary parts at the IFFT output.

is moved to a passband region including the one from 0 to $+2B$ on the positive axis and the other from $-2B$ to 0 on the negative axis. This gives rise to $f_{\mathrm{c}} = B$. For discrete-time upconversion, the equivalent carrier frequency is:

$$\hat{f}_{\mathrm{c}} = \frac{B}{2f_{\mathrm{s}}}. \tag{7}$$

The upconverted SSB-OFDM signal can be expressed as:

$$\tilde{x}_{\mathrm{SSB}}(n) = \tilde{x}_{\mathrm{i}}(n)\cos\left(2\pi\hat{f}_{\mathrm{c}}n\right) \pm \tilde{x}_{\mathrm{q}}(n)\sin\left(2\pi\hat{f}_{\mathrm{c}}n\right), \tag{8}$$

with plus and minus signs leading to LSSB and USSB cases, respectively. After the D/A conversion, the signal power is scaled and clipped based on the LED dynamic range and the required DC bias. The D/A converter needs to provide $2f_{\mathrm{s}}$ samples/s. The continuous-time SSB-OFDM waveform with DC bias is:

$$x_{\mathrm{SSB}}(t) = \sqrt{\frac{P_{\mathrm{elec}}}{\xi_{\mathrm{SSB}}}}\tilde{x}_{\mathrm{SSB}}(t) + x_{\mathrm{DC}}. \tag{9}$$

The receiver performs an inverse function of the transmitter to recover the baseband SSB-OFDM signal, thus the receiver processing is skipped for brevity.

An illustrative example helps clarify the presented analysis. Accordingly, the end-to-end transceiver system of SSB-OFDM is simulated and the results are shown in Figs. 3–5. The effect of VLC system is modeled as a bandlimited and flat channel of bandwidth $B = 20$ MHz. Apart from this, an ideal lossless and noiseless channel is assumed to verify the functionality of the transceiver. For Monte-Carlo simulations, a random sequence of information bits is generated and modulated per subcarrier using 16-QAM. An IFFT/FFT of length $N = 1024$ and an electrical power of $P_{\mathrm{elec}} = 1$ W are considered.

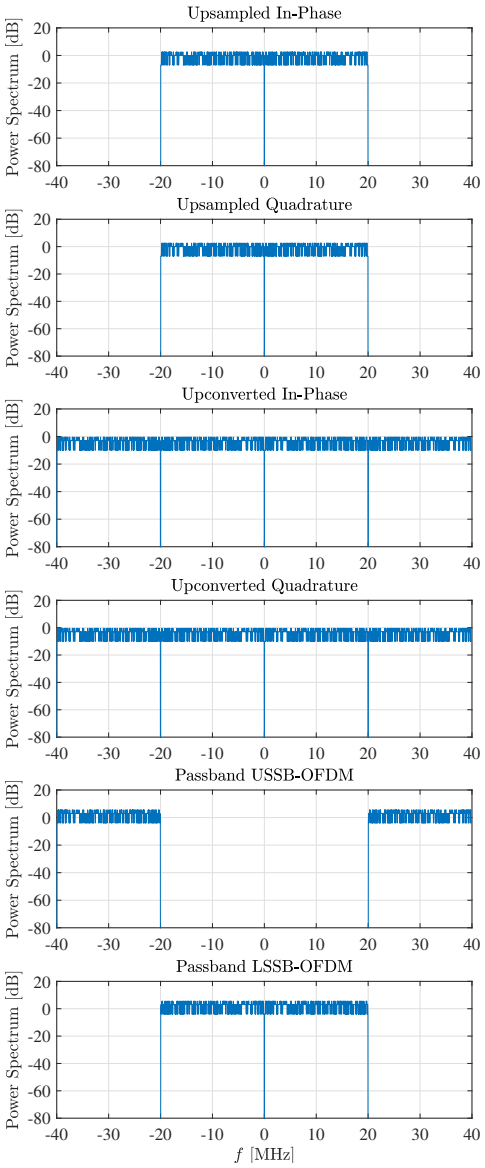


Fig. 4: Power spectra of the in-phase and quadrature signals after upsampling and upconverstion at the transmitter, and the resulting passband LSSB/USSB-OFDM signals.

IV. Performance Analysis

In the following, the SE performance of SSB-OFDM is assessed and compared with DCO-OFDM. In addition, its BER performance over an additive white Gaussian noise (AWGN) channel is studied by evaluating the received electrical SNR.

A. Spectral Efficiency

The SE of a communication system is defined as the realized bit rate divided by the used bandwidth. With the single-sided bandwidth, B, of the VLC system, both the considered OFDM schemes have a baseband bandwidth of $2B$. The corresponding OFDM symbol rate is equal to $\frac{2B}{N}$ symbols/s. Consequently, by transmitting on N_{QAM} subcarriers, the data rate becomes:

$$R = \frac{2B}{N} N_{\mathrm{QAM}} \log_2 M \quad \text{bits/s}, \tag{10}$$

Authorized licensed use limited to: University of Wollongong. Downloaded on August 14,2020 at 14:36:50 UTC from IEEE Xplore. Restrictions apply.

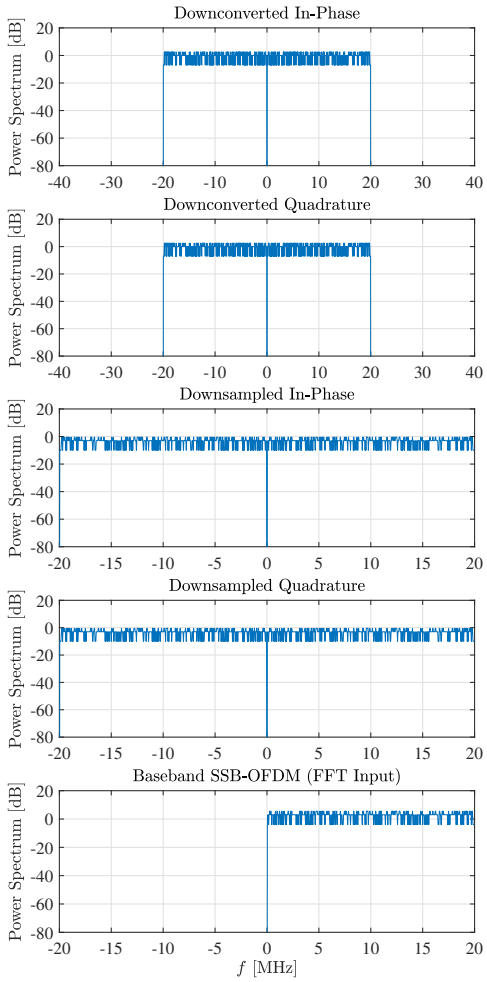


Fig. 5: Power spectra of the in-phase and quadrature signals after downconversion and downsampling of LSSB-OFDM at the receiver, and the recovered basedband SSB-OFDM signal.

and a unified expression of the SE is obtained as follows:

$$\eta = \frac{R}{B} = \frac{2N_{\mathrm{QAM}}}{N}\log_2 M \quad \text{bits/s/Hz}. \tag{11}$$

Both cases of DCO-OFDM and DC-biased LSSB-OFDM have $N_{\mathrm{QAM}} = \frac{N}{2} - 1$. Hence, they achieve an equal SE of:

$$\eta_{\mathrm{DCO}} = \eta_{\mathrm{SSB}} = \frac{N-2}{N}\log_2 M. \tag{12}$$

B. Received SNR

In VLC systems, the frequency response of the free space propagation channel is almost flat near DC, so it can be solely characterized by the DC gain, denoted by H_0, which relates the transmitted and received average optical powers, represented as P_{t} and P_{r}, respectively, via $P_{\mathrm{r}} = H_0 P_{\mathrm{t}}$ [17]. This causes an attenuation for the received optical power, depending on the link configuration. After the optical-to-electrical conversion, the detected photocurrent is of the form $R_{\mathrm{PD}}P_{\mathrm{r}} = R_{\mathrm{PD}}H_0P_{\mathrm{t}}$, using R_{PD} to denote the PD responsivity. The photodetection is subject to shot noise and thermal noise terms, affecting the

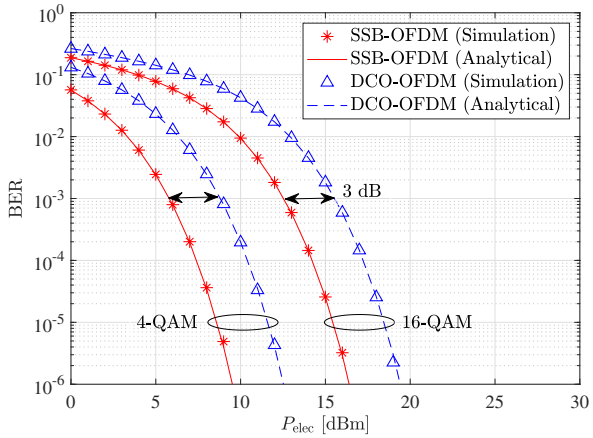


Fig. 6: Comparison of the BER performance of DC-biased SSB-OFDM with that of DCO-OFDM.

received constellation. The aggregate noise at the receiver is modeled as a zero-mean, complex-valued AWGN with a two-sided power spectral density (PSD) of $\frac{N_0}{2}$ per dimension. The noise variance is equal to $\sigma_{\mathrm{n}}^2 = B_{\mathrm{n}}N_0$, where $B_{\mathrm{n}} = \frac{2N_{\mathrm{QAM}}}{N}B$ is the equivalent one-sided noise bandwidth. For both cases of DCO-OFDM and SSB-OFDM, it follows that $B_{\mathrm{n}} = \frac{N-2}{N}B$. Therefore, the received electrical SNR per symbol for both schemes can be expressed in a unified manner by using:

$$\gamma_{\mathrm{s}} = \frac{N R_{\mathrm{PD}}^2 H_0^2 P_{\mathrm{t}}^2}{(N-2)\,B N_0}. \tag{13}$$

Based on (3) and (9), the total electrical transmission power for DCO-OFDM and SSB-OFDM, respectively, are $P_{\mathrm{t}}^2 = \frac{P_{\mathrm{elec}}}{\xi_{\mathrm{DCO}}}$ and $P_{\mathrm{t}}^2 = \frac{P_{\mathrm{elec}}}{\xi_{\mathrm{SSB}}}$. By substituting these parameters into (13), the symbol SNR for each scheme can be derived as follows:

$$\gamma_{\mathrm{DCO}} = \frac{R_{\mathrm{PD}}^2 H_0^2 P_{\mathrm{elec}}}{\xi_{\mathrm{DCO}}^2 B N_0}, \tag{14}$$

$$\gamma_{\mathrm{SSB}} = \frac{2R_{\mathrm{PD}}^2 H_0^2 P_{\mathrm{elec}}}{\xi_{\mathrm{DCO}}^2 B N_0}, \tag{15}$$

where $\xi_{\mathrm{DCO}} = 2\xi_{\mathrm{SSB}} = \frac{N-2}{N}$. It follows that $\gamma_{\mathrm{SSB}} = 2\gamma_{\mathrm{DCO}}$. This SNR gain of 2 is the result of allocating the electrical power to one half of an SSB-OFDM frame, whereas the same power is assigned to the whole frame in DCO-OFDM because of the Hermitian symmetry constraint.

C. BER Performance

In [18], the exact BER expression of an M-ary square QAM constellation in AWGN has been derived as a summation of M terms. A tight approximation can be extracted by keeping the first two terms and neglecting the remainder. As a result, an analytical expression for the BER performance of M-QAM OFDM-based VLC systems is obtained as:

$$\mathrm{BER} = \frac{4(\sqrt{M}-1)}{\sqrt{M}\log_2 M} Q\left(\sqrt{\frac{3}{M-1}\gamma_{\mathrm{s}}}\right) + \frac{4(\sqrt{M}-2)}{\sqrt{M}\log_2 M} Q\left(3\sqrt{\frac{3}{M-1}\gamma_{\mathrm{s}}}\right), \tag{16}$$

Authorized licensed use limited to: University of Wollongong. Downloaded on August 14,2020 at 14:36:50 UTC from IEEE Xplore. Restrictions apply.

where $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-\frac{u^2}{2}} du$ is the Gaussian Q-function; and $\gamma_{\rm s}$ is the received electrical SNR per symbol. The BER for DCO-OFDM and SSB-OFDM schemes is readily deduced from (16) upon replacing $\gamma_{\rm s}$ by their corresponding SNR from among (14) and (15).

The BER performance of DCO-OFDM and SSB-OFDM is studied based on numerical results for both the presented analysis and Monte-Carlo simulations. For each case, the end-to-end transceiver chain is simulated and the BER results are compared with those obtained by the analytical expression in (16). Monte-Carlo simulations are conducted by averaging the results over random realizations for 10^4 OFDM frames. Each frame encompasses $N = 1024$ subcarriers, of which 511 subcarriers are QAM-modulated. The VLC system bandwidth is $B = 20$ MHz. A typical indoor VLC channel is assumed, with an LED light fixture installed on the ceiling and a receiver device located on a table hight above the floor. A channel DC gain of $H_0 = 10^{-5}$, a PD responsivity of $R_{\rm PD} = 0.5$ A/W, and a noise PSD of $N_0 = 10^{-21}$ A^2/Hz are used, which are all borrowed from [17]. Furthermore, the electrical power, $P_{\rm elec}$, is varied in a range from 0 dBm to 30 dBm, so by assuming a maximum allowed optical power of about 35 dBm, it is ensured that no significant clipping distortion occurs when choosing an appropriate DC bias.

Fig. 6 illustrates the BER results for two cases of 4-QAM and 16-QAM. In either case, it can be seen that there is a 3 dB gap whereby SSB-OFDM performs better than DCO-OFDM, verifying the same observation made by the SNR analysis, during which a relative SNR gain of 2 was predicted. Besides, the analytical and simulation results are in excellent match, which confirms the accuracy of the approach. The SNR gain can be exploited to choose a higher modulation order to increase the data rate for the same BER performance, or to achieve a lower BER level for the same data rate. This shows the potential advantage of employing SSB-OFDM for band-limited VLC systems. However, the computational complexity for SSB-OFDM is higher than DCO-OFDM due the additional DSP operations required, including upsampling/downsampling and multiplications used for upconversion/downconversion.

V. Conclusions

The application of SSB-OFDM for optical IM-DD systems was investigated. The problem of the bandwidth limitation in VLC primarily makes the transmission of a bandpass signal challenging. To tackle this challenge, step-by-step processing for the baseband to passband conversion of an SSB-OFDM signal was presented until the point where its spectrum fits within the low-pass frequency response of a VLC system. It was shown that this type of transmission yields the same spectral efficiency of a DCO-OFDM scheme when using an equal number of data-carrying subcarriers and DC biasing. The absence of Hermitian symmetry in SSB-OFDM led to a 3 dB gain in the electrical power with respect to DCO-OFDM. It can be used to realize a higher constellation size based on rate-adaptive modulations, or as a means to enhance the detection performance in error-sensitive applications. This comes at the cost of an increased complexity in the DSP implementation. Future work involves the incorporation of system nonidealities and impairments such as frequency selectivity and nonlinear clipping into an extended analysis of SSB-OFDM for VLC.

Acknowledgment

Prof. Harald Haas acknowledges financial support from the Engineering and Physical Sciences Research Council (EPSRC) under the Established Career Fellowship grant EP/R007101/1 as well as the Wolfson Foundation and the Royal Society. All authors acknowledge financial support from EPSRC under program grant EP/S016570/1 'Terabit Bidirectional Multi-User Optical Wireless System (TOWS) for 6G LiFi'.

References

[1] J. Vucic, C. Kottke, S. Nerreter, K. D. Langer, and J. W. Walewski, "513 Mbit/s Visible Light Communications Link Based on DMT-Modulation of a White LED," *IEEE/OSA J. Lightw. Technol.*, vol. 28, no. 24, pp. 3512–3518, Dec. 2010.

[2] H. Elgala, R. Mesleh, and H. Haas, "Practical Considerations for Indoor Wireless Optical System Implementation using OFDM," in *Proc. 10th Int. Conf. Telecommun. (ConTEL)*, Jun. 2009, pp. 25–29.

[3] J. Armstrong, "OFDM for Optical Communications," *IEEE/OSA J. Lightw. Technol.*, vol. 27, no. 3, pp. 189–204, Feb. 2009.

[4] O. Gonzalez, R. Perez-Jimenez, S. Rodriguez, J. Rabadan, and A. Ayala, "OFDM over indoor wireless optical channel," *IEE Proceedings - Optoelectronics*, vol. 152, no. 4, pp. 199–204, Aug. 2005.

[5] J. Armstrong and A. J. Lowery, "Power efficient optical OFDM," *IET Electron. Lett.*, vol. 42, no. 6, pp. 370–372, Mar. 2006.

[6] A. J. Lowery and J. Armstrong, "Orthogonal-frequency-division multiplexing for dispersion compensation of long-haul optical systems," *OSA Opt. Express*, vol. 14, no. 6, pp. 2079–2084, Mar. 2006.

[7] B. J. C. Schmidt, A. J. Lowery, and J. Armstrong, "Experimental Demonstrations of Electronic Dispersion Compensation for Long-Haul Transmission Using Direct-Detection Optical OFDM," *IEEE/OSA J. Lightw. Technol.*, vol. 26, no. 1, pp. 196–203, Jan. 2008.

[8] H. Kim, D. Kim, S. Yang, Y. Son, and S. Han, "Single side-band orthogonal frequency division multiplexing signal transmission in RF carrier allocated visible light communication," *IET Optoelectronics*, vol. 7, no. 6, pp. 125–130, Dec. 2013.

[9] ——, "Mitigation of Inter-Cell Interference Utilizing Carrier Allocation in Visible Light Communication System," *IEEE Commun. Lett.*, vol. 16, no. 4, pp. 526–529, Apr. 2012.

[10] ——, "An Indoor Visible Light Communication Positioning System Using a RF Carrier Allocation Technique," *IEEE/OSA J. Lightw. Technol.*, vol. 31, no. 1, pp. 134–144, Jan. 2013.

[11] A. M. Khalid, G. Cossu, R. Corsini, P. Choudhury, and E. Ciaramella, "1-Gb/s Transmission Over a Phosphorescent White LED by Using Rate-Adaptive Discrete Multitone Modulation," *IEEE Photon. J.*, vol. 4, no. 5, pp. 1465–1473, Oct. 2012.

[12] D. Dardari, V. Tralli, and A. Vaccari, "A Theoretical Characterization of Nonlinear Distortion Effects in OFDM Systems," *IEEE Trans. Commun.*, vol. 48, no. 10, pp. 1755–1764, Oct. 2000.

[13] J. Armstrong and B. J. C. Schmidt, "Comparison of Asymmetrically Clipped Optical OFDM and DC-Biased Optical OFDM in AWGN," *IEEE Commun. Lett.*, vol. 12, no. 5, pp. 343–345, May 2008.

[14] J. G. Proakis and M. Salehi, *Communication Systems Engineering*, 2nd ed. Prentice-Hall, 2002.

[15] D. J. F. Barros and J. M. Kahn, "Comparison of Orthogonal Frequency-Division Multiplexing and On-Off Keying in Amplified Direct-Detection Single-Mode Fiber Systems," *IEEE/OSA J. Lightw. Technol.*, vol. 28, no. 12, pp. 1811–1820, Jun. 2010.

[16] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Pearson, 2010.

[17] T. Komine and M. Nakagawa, "Fundamental Analysis for Visible-Light Communication System using LED Lights," *IEEE Trans. Consum. Electron.*, vol. 50, no. 1, pp. 100–107, Feb. 2004.

[18] K. Cho and D. Yoon, "On the General BER Expression of One- and Two-Dimensional Amplitude Modulations," *IEEE Trans. Commun.*, vol. 50, no. 7, pp. 1074–1080, 2002.

Authorized licensed use limited to: University of Wollongong. Downloaded on August 14,2020 at 14:36:50 UTC from IEEE Xplore. Restrictions apply.